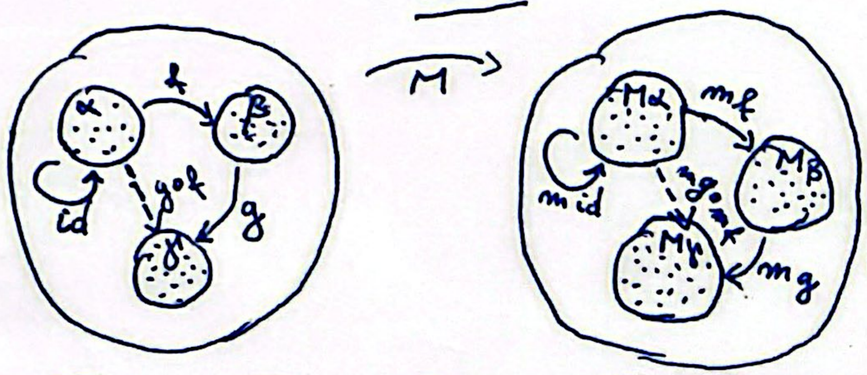
MONAD

C.T.



„objects"

„Mathlib. Category Theory. Monad"

- $\text{pure} = \text{pure}_\alpha : \alpha \to M\alpha$ „unit"
- $m : (\alpha \to \beta) \to (M\alpha \to M\beta)$ „map" „morphisms" „fmap"
- $\text{join} = \text{join}_\alpha : MM\alpha \to M\alpha$ „multiplication"

„(endo) functor"

$$\begin{cases} \bullet\ (m\ id)\ x' = x' \\ \bullet\ m(g \circ f)\ x' = (m\ g \circ m\ f)\ x' \end{cases}$$

$$\begin{cases} \bullet\ (m\ f)(\text{pure}\ x) = \text{pure}\ (f\ x) \\ \bullet\ \text{join}\ (m(m\ f)\ x'') = m\ f\ (\text{join}\ x'') \end{cases}$$

- $\text{join}\ (m\ \text{join}\ x''') = \text{join}\ (\text{join}\ x''')$
- $\text{join}\ ((m\ \text{pure})\ x') = \text{join}\ (\text{pure}\ x') = x'$

„natural transformations"

$$\begin{tikzcd} \alpha \arrow[r, "\text{pure}"] \arrow[d, "f"'] & M\alpha \arrow[d, "mf"] \\ \beta \arrow[r, "\text{pure}"] & M\beta \end{tikzcd}$$

$$\begin{tikzcd} MM\alpha \arrow[r, "\text{join}"] \arrow[d, "m(mf)"'] & M\alpha \arrow[d, "mf"] \\ MM\beta \arrow[r, "\text{join}"] & M\beta \end{tikzcd}$$

$$\begin{tikzcd} MMM\alpha \arrow[r, "\text{join}_{M\alpha}"] \arrow[d, "m\,\text{join}_\alpha"'] & MM\alpha \arrow[d, "\text{join}_\alpha"] \\ MM\alpha \arrow[r, "\text{join}_\alpha"] & M\alpha \end{tikzcd}$$

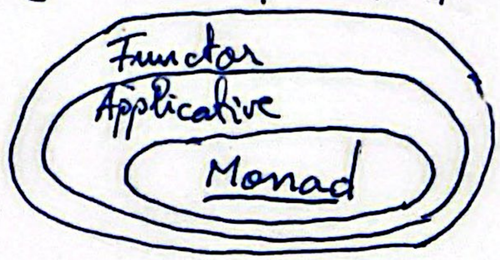
„associative"

$$\begin{tikzcd} M\alpha \arrow[r, "\text{pure}_{M\alpha}"] \arrow[d, "m\,\text{pure}_\alpha"'] \arrow[dr, "id"] & MM\alpha \arrow[d, "\text{join}"] \\ MM\alpha \arrow[r, "\text{join}"] & M\alpha \end{tikzcd}$$

„unity"

F.P.

E.g. $M = \text{Option}$; $\text{Option}\ \alpha = \{\text{none}\} \cup \{\text{some}\ x \mid x : \alpha\}$



„Init. Prelude/Control"

- $\text{pure} = \text{pure}_\alpha : \alpha \to M\alpha$ „return"
- $>>= : M\alpha \to (\alpha \to M\beta) \to M\beta$ „bind"

- $x' >>= \text{pure} = x'$
- $(\text{pure}\ x) >>= f' = f'\ x'$
- $x' >>= f' >>= g' = x' >>= (\text{fun}\ x \Rightarrow f'\ x >>= g')$

$(m\ f)\ x' := x' >>= \text{pure} \circ f$

$\text{join}\ x'' := x'' >>= id_{M\alpha}$

$x' >>= f' := \text{join}\ ((m\ f')\ x')$